

The screenshot shows the ChatPDF web interface. On the left is a sidebar with a 'New Chat' button, a 'New Folder' button, and a list of PDF files: 'IN14.pdf', '3316481.pdf', and 'wirelessen.pdf'. Below the list is a 'Sign In' button and an 'AI Scholar' button. The main area displays the PDF document '3316481.pdf'. The document text is partially visible, discussing blockchain security and privacy. On the right is a 'Chat' window with a greeting: 'Hello and welcome! We're excited to share insights from our PDF on "Security and Privacy on Blockchain." This document explores the innovative world of blockchain technology, highlighting its applications in cryptocurrency, smart contracts, and more. It delves into the essential security properties that underpin these systems and discusses the ongoing challenges related to security and privacy in various blockchain applications. We hope you find this overview informative and engaging! Here are three example questions you can ask about the file:'. Below the greeting are three suggested questions in a list: 'What are the basic security properties of blockchain technology?', 'How does blockchain enhance trust in online transactions?', and 'What are some of the privacy concerns associated with blockchain applications?'. At the bottom right of the chat window is a button labeled 'problem statement in this paper'.

Figure 5: Finding a problem statement using ChatPDF

The screenshot shows the Perplexity AI web interface. The top bar displays the URL 'perplexity.ai/search/key-points-and-main-work-in-th-07CDadGuTBCLfaer0JAjhg'. Below the URL bar is a search bar with the text 'key points and main work in this paper'. The main content area is titled 'Key Points' and lists three points: '1. Diversity of Wireless Technologies: The paper highlights the variety of wireless technologies, including WPANs (like Bluetooth), 3G cellular networks (like CDMA2000 and UMTS), and WLANs (IEEE Std. 802.11). Each technology has unique coverage and bandwidth characteristics, leading to the necessity for interworking among them to achieve high-performance wireless communications.', '2. Security Challenges: A significant focus is placed on the differing security architectures of 3G and WLAN systems. The authors discuss how each system faces distinct security issues, such as fraudulent access in 3G systems and vulnerabilities in WLAN's original security architecture (WEP), which led to the development of WPA.', and '3. Interworking Security Issues: The paper outlines how integrating 3G and WLAN systems presents security challenges due to conflicting security assumptions that may compromise overall security. It emphasizes the need for efficient authentication mechanisms during handover between networks to prevent service disruption.' On the left side of the interface is a sidebar with a 'New Thread' button, a search bar, and a list of categories: 'Home', 'Discover', 'Spaces', and 'Library'. Below the 'Library' category is a list of items: 'key points and main work in', 'generate 5 mcqs from this p', 'main work in this document', and 'main theme of this paper in'.

Figure 6: Extraction of deep content from an uploaded file using perplexity.ai

professional development, customised teaching materials, interactive learning, plagiarism detection and language support.

It helps researchers with literature reviews, generation of citations, writing grant proposals, hypothesis generation, trend analysis, data analysis support, technical writing assistance, code assistance, and multidisciplinary insights.

Perplexity has an in-built feature using which the relevant research questions are automatically identified from the uploaded documents. In this way, it becomes easy to check which specific queries are associated with a document. In parallel to querying a document, the questions can also be asked in a direct way so that the answers can be fetched accurately from its own model along with the source of the content.

By using AI based software applications, teachers as well as researchers can search the content from uploaded documents so that academic and research tasks can be completed within time and with accuracy. To work with such AI based tools, however, ethical values must be followed so that the content aids in understanding concepts and enhancing knowledge. **END** 🐼

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